

Physical scale priors and starting values

$$\pi(\boldsymbol{\theta}), \boldsymbol{\theta}^{(0)}$$

Transform to latent Gaussian scale

$$z_k \sim N(0, 1), z_k^{*(0)} = T_k^{-1}(\boldsymbol{\theta}_k^{(0)})$$

*Iterative inlabru
approximation*

Linearise ETAS predictors $\eta_r(\mathbf{z})$ at $\mathbf{z}^{*(s)}$

$$\log \Lambda_0, \log \Lambda_{i,h}, \log \lambda_j$$

Form local latent Gaussian model

surrogate Poisson representation

INLA fit

candidate joint mode $\hat{\mathbf{z}}^{(s)}$

approximate marginal posteriors $\tilde{\pi}^{(s)}(z_k | \mathbf{y})$

Line search and update

next linearisation point $\mathbf{z}^{*(s+1)}$

No

Converged?

Yes

Final INLA fit at converged linearisation point

approximate marginal posteriors $\tilde{\pi}(z_k | \mathbf{y})$

Transform to physical ETAS scale

$$\theta_k = T_k(z_k) \rightarrow \tilde{\pi}(\theta_k | \mathbf{y})$$